

Math 521, Section 2
Practice Midterm 2
Spring 2026

First name: _____ **Last name:** _____

Instructions:

- This exam contains 4 problems, and there are a total of 40 points available.
- Show all your work in the space provided. You may also use the backs of pages.
- No outside resources are allowed, including notes, calculators, textbooks, etc.

Question	Points	Score
1	10	
2	10	
3	10	
4	10	
Total:	40	

1. (10 points) Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely and $\{b_n\}_{n \geq 1}$ is a bounded sequence, then the series $\sum_{n=1}^{\infty} a_n b_n$ converges.

2. For each of the following series, determine if it converges and prove your answer.

(a) (5 points) $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n^n}}$

(b) (5 points) $\sum_{n=1}^{\infty} \left(\frac{1}{\sqrt{n}} - \frac{1}{\sqrt{n+1}} \right)$

3. Suppose that d and $\tilde{d}$ are two metrics on $\mathbb{R}$ such that

$$\tilde{d}(x, y) \leq d(x, y) \quad \text{for all } x, y \in \mathbb{R}.$$

Prove or disprove the following statements:

- (a) (5 points) For any $r > 0$, we have $\{x \in \mathbb{R} : \tilde{d}(x, 0) < r\} \subseteq \{x \in \mathbb{R} : d(x, 0) < r\}$.
(b) (5 points) For any $r > 0$, we have $\{x \in \mathbb{R} : \tilde{d}(x, 0) < r\} \supseteq \{x \in \mathbb{R} : d(x, 0) < r\}$.

4. (10 points) Recall that

$$d_1(x, y) = |x_1 - y_1| + |x_2 - y_2|$$

is a metric on $\mathbb{R}^2$. Suppose that $x^n = (x_1^n, x_2^n)$ is a sequence of points in $\mathbb{R}^2$. Prove that the sequence $\{x^n\}_{n \geq 1}$ converges in $(\mathbb{R}^2, d_1)$ if and only if the sequences $\{x_1^n\}_{n \geq 1}$ and $\{x_2^n\}_{n \geq 1}$ of real numbers both converge.

Extra paper

1.) We'll use the Cauchy criterion. Fix $\varepsilon > 0$.
As $\{b_n\}$ is bounded, $\exists M > 0$ s.t.

$$|b_n| \leq M \quad \forall n \geq 1.$$

As $\sum |a_n|$ converges, $\exists N \in \mathbb{N}$ s.t.

$$\sum_{k=m}^n |a_k| < \frac{\varepsilon}{M} \quad \forall n \geq m \geq N.$$

So, for $n \geq m \geq N$ we have

$$\begin{aligned} \left| \sum_{k=m}^n a_k b_k \right| &\leq \sum_{k=m}^n |a_k| \cdot |b_k| \\ &\leq M \sum_{k=m}^n |a_k| < M \cdot \frac{\varepsilon}{M} = \varepsilon. \end{aligned}$$

Therefore $\sum a_n b_n$ converges by the Cauchy criterion. $\square$

2.) (a.) Set $a_n = \frac{1}{\sqrt{n^n}}$. Then

$$|a_n|^{\frac{1}{n}} = \left(\frac{1}{n^{\frac{n}{2}}} \right)^{\frac{1}{n}} = \frac{1}{n^{\frac{1}{2}}} = \sqrt{\frac{1}{n}} \rightarrow 0 \text{ as } n \rightarrow \infty,$$

and so $\limsup |a_n|^{\frac{1}{n}} = 0 < 1$. Therefore the series

converges by the root test.

(b.) Set $s_n = \sum_{k=1}^n \left(\frac{1}{\sqrt{k}} - \frac{1}{\sqrt{k+1}} \right)$.

Claim: $s_n = 1 - \frac{1}{\sqrt{n+1}}$ $\forall n \geq 1$. We'll argue by induction.

• Base case: For $n=1$, we have $s_1 = \frac{1}{\sqrt{1}} - \frac{1}{\sqrt{2}} = 1 - \frac{1}{\sqrt{2}}$.

• Inductive step: Suppose $s_n = 1 - \frac{1}{\sqrt{n+1}}$ for some $n \geq 1$. Then

$$\begin{aligned} s_{n+1} &= s_n + \left(\frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}} \right) = 1 - \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+1}} - \frac{1}{\sqrt{n+2}} \\ &= 1 - \frac{1}{\sqrt{n+2}} \end{aligned}$$

So the formula holds for $n+1$.

Together, we conclude $s_n = 1 - \frac{1}{\sqrt{n+1}} \quad \forall n \geq 1$ by induction.

We know $\frac{1}{n} \rightarrow 0$ as $n \rightarrow \infty$ from lecture, and so the subsequence $\frac{1}{n+1} \rightarrow 0$ as well. So, by the limit laws,

$$s_n = 1 - \frac{1}{\sqrt{n+1}} = 1 - \sqrt{\frac{1}{n+1}} \rightarrow 1 \quad \text{as } n \rightarrow \infty.$$

So the series converges to 1. □

3.) (a.) Disprove: Consider $d(x, y) = |x - y|$, $\tilde{d}(x, y) = \min\{d(x, y), 1\}$, and $r = 2$. Then

$$\{x \in \mathbb{R} : \tilde{d}(x, 0) < 2\} = \mathbb{R}$$

$$\{x \in \mathbb{R} : d(x, 0) < 2\} = (-2, 2)$$

and $\mathbb{R} \not\subset (-2, 2)$.

(b.) Prove: Fix $r > 0$. Then

$$y \in \{x \in \mathbb{R} : d(x, 0) < r\} \Rightarrow d(y, 0) < r$$

$$\Rightarrow \tilde{d}(y, 0) \leq d(y, 0) < r$$

$$\Rightarrow y \in \{x \in \mathbb{R} : \tilde{d}(x, 0) < r\}. \quad \square$$

4.) $\Rightarrow$: Suppose $x^n \rightarrow x$ in $(\mathbb{R}^2, d_1)$. Then $d_1(x^n, x) \rightarrow 0$ as $n \rightarrow \infty$. Note that

$$0 \leq |x_1^n - x_1| \leq |x_1^n - x_1| + |x_2^n - x_2| = d_1(x^n, x) \quad \forall n \geq 1.$$

So $\lim_{n \rightarrow \infty} x_1^n = x_1$ by the squeeze theorem. Similarly,

$$0 \leq |x_2^n - x_2| \leq |x_1^n - x_1| + |x_2^n - x_2| = d_1(x^n, x),$$

and so $\lim_{n \rightarrow \infty} x_2^n = x_2$.

$\Leftarrow$: Suppose $\lim_{n \rightarrow \infty} x_1^n = x_1$ and $\lim_{n \rightarrow \infty} x_2^n = x_2$.

Then $|x_1^n - x_1| \rightarrow 0$ and $|x_2^n - x_2| \rightarrow 0$ as $n \rightarrow \infty$, and so by the limit law for addition,

$$|x_1^n - x_1| + |x_2^n - x_2| \rightarrow 0 \quad \text{as } n \rightarrow \infty.$$

Therefore $d_1(x^n, x) \rightarrow 0$ as $n \rightarrow \infty$, and so $x^n \rightarrow x$ in $(\mathbb{R}^2, d_1)$. $\square$

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Total:	40	

1. (10 points) Let $\{a_n\}_{n \geq 1}$ be a sequence of positive real numbers. Prove that $\lim_{n \rightarrow \infty} a_n = \infty$ if and only if $\lim_{n \rightarrow \infty} \frac{1}{a_n} = 0$.

2. (10 points) Let $\sum_{n=1}^{\infty} a_n$ be a series of real numbers. Prove that if $\sum_{n=1}^{\infty} a_n$ converges absolutely, then $\sum_{n=1}^{\infty} a_{2n}$ converges.

3. (10 points) Let (X, d) be a metric space and let A be a nonempty subset of X . Prove that A is open in (X, d) if and only if there exists a set I , positive numbers $r_i > 0$, and points $a_i \in X$ so that $A = \bigcup_{i \in I} B_{r_i}(a_i)$.

4. Recall that $d_2(x, y) = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2}$ is a metric on $\mathbb{R}^2$. Consider the subspace

$$Y = \{(x, y) \in \mathbb{R}^2 : y > 0\}$$

with the induced metric d_Y . **Prove or disprove** the following statements:

- (a) (5 points) If $A \subseteq Y$ is open in (Y, d_Y) , then A is open in $(\mathbb{R}^2, d_2)$.
- (b) (5 points) If $B \subseteq Y$ is closed in (Y, d_Y) , then B is closed in $(\mathbb{R}^2, d_2)$.

Extra paper

1.) $\Rightarrow$: Fix $\varepsilon > 0$. As $\lim a_n = \infty$, then $\exists N \in \mathbb{N}$ s.t.

$$a_n > \frac{1}{\varepsilon} \quad \forall n \geq N.$$

So, for $n \geq N$,

$$|\frac{1}{a_n} - 0| = \frac{1}{a_n} < \varepsilon$$

Therefore $\lim \frac{1}{a_n} = 0$.

$\Leftarrow$: Fix $M > 0$. As $\lim \frac{1}{a_n} = 0$, then $\exists N \in \mathbb{N}$ s.t.

$$0 < \frac{1}{a_n} = |\frac{1}{a_n} - 0| < \frac{1}{M} \quad \forall n \geq N$$

So, for $n \geq N$,

$$a_n > M$$

Therefore $\lim a_n = \infty$. □

2.) Fix $\varepsilon > 0$. As $\sum |a_n|$ converges, then by the Cauchy criterion, $\exists N \in \mathbb{N}$ s.t.

$$\sum_{k=m}^n |a_k| < \varepsilon \quad \forall n \geq m \geq N.$$

So, for $n \geq m \geq N$, we have

$$\left| \sum_{k=m}^n a_{2k} \right| \leq \sum_{k=m}^n |a_{2k}| \leq \sum_{l=2m}^{2n} |a_l| < \varepsilon$$

↑
since $2n \geq 2m \geq 2N \geq N$.

Therefore $\sum a_{2n}$ converges by the Cauchy criterion. □

3.) $\Rightarrow$: Suppose $A \subseteq X$ is open. Then $\forall a \in A$

$\exists r_a > 0$ s.t. $B_{r_a}(a) \subseteq A$. We claim that

$$A = \bigcup_{a \in A} B_{r_a}(a),$$

which will show that $I=A$ is a set that works and so finish the proof. Note that

$$x \in A \Rightarrow x \in B_{r_x}(x) \subseteq A \Rightarrow x \in \bigcup_{a \in A} B_{r_a}(a),$$

and so $A \subseteq \bigcup_{a \in A} B_{r_a}(a)$. Conversely,

$$\begin{aligned} y \in \bigcup_{a \in A} B_{r_a}(a) &\Rightarrow \exists a \in A \text{ s.t. } y \in B_{r_a}(a) \\ &\Rightarrow y \in B_{r_a}(a) \subseteq A \Rightarrow y \in A, \end{aligned}$$

and so $A \supseteq \bigcup_{a \in A} B_{r_a}(a)$. Together, we conclude

$$A = \bigcup_{a \in A} B_{r_a}(a), \text{ as desired.}$$

$\Leftarrow$: Suppose $A = \bigcup_{i \in I} B_{r_i}(a_i)$ for some set I , numbers $r_i > 0$, and points $a_i \in X$. From lecture, we know $B_{r_i}(a_i)$ is open $\forall i \in I$. So A is open, since the arbitrary union of open sets is open. $\square$

4.) (a.) Prove: By lecture, it suffices to show that Y is open in $(\mathbb{R}^2, d_2)$. Fix $(x, y) \in Y$. Then $y > 0$, and $B_y((x, y)) \subseteq Y$ since

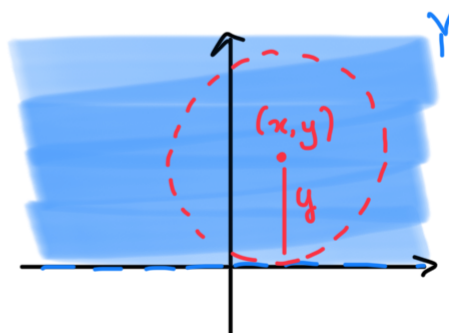
$$(x', y') \in B_y(x, y) \Rightarrow \sqrt{|x-x'|^2 + |y-y'|^2} < y$$

$$\Rightarrow |y-y'| \leq \sqrt{|x-x'|^2 + |y-y'|^2} < y$$

$$\Rightarrow y' > y - (y-y') > y - y = 0$$

$$\Rightarrow (x', y') \in Y.$$

So Y is open.



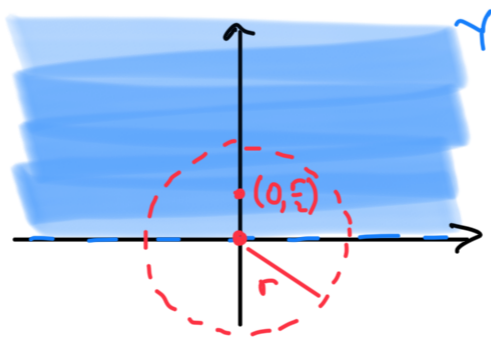
(b.) Disprove: Consider $B=Y$. Then

- Y is closed in (Y, d_Y) (this is true for any metric space)

- Y is not closed in $(\mathbb{R}^2, d_2)$: Note that $(0, 0) \in \overline{Y}^{\mathbb{R}^2}$, since given $r > 0$, the point $(0, \frac{r}{2})$ is in Y and $B_r^{\mathbb{R}^2}((0, 0))$. So

$\overline{Y}^{\mathbb{R}^2} \neq Y$, and thus Y is not closed in $(\mathbb{R}^2, d_2)$.

□



Math 521, Section 2
Practice Midterm 1
Spring 2026

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1	10	
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Total:	40	

1. (10 points) Let $(F, +, \cdot)$ be a field. Prove that if $a, b \in F \setminus \{0\}$, then $a^{-1} \cdot b^{-1} = (ab)^{-1}$.

Make sure that each step follows from a property of fields that we stated in lecture. You do not need to cite each fact by name; just show your logic one step at a time.

2. (a) (3 points) State the definition of: the set A is *countable*.
(b) (7 points) Prove that the set

$$B = \{(x, y) \in \mathbb{R}^2 : x \in \mathbb{N} \text{ and } y \in \{0, 1\}\}$$

satisfies the definition in part (a).

3. (10 points) Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets, and define

$$S = \{a + b : a \in A \text{ and } b \in B\}.$$

Prove that $\sup S = \sup A + \sup B$.

4. Let $\{a_n\}_{n \geq 1}$ be a sequence and $a \in \mathbb{R}$. **Prove or disprove** the following statements:
- (a) (5 points) If $\{a_n\}_{n \geq 1}$ converges to a , then the sequence $\{|a_n|\}_{n \geq 1}$ converges to $|a|$.
 - (b) (5 points) If the sequence $\{|a_n|\}_{n \geq 1}$ converges to $|a|$, then $\{a_n\}_{n \geq 1}$ converges to a .

Extra paper

1.) (This is HW3#1e.)

Fix $a, b \neq 0$. Then

$$\begin{aligned}(a^{-1} \cdot b^{-1}) \cdot (ab) &= (a^{-1} \cdot b^{-1}) \cdot (ba) \\&= (a^{-1} \cdot (b^{-1} \cdot b)) \cdot a \\&= (a^{-1} \cdot 1) \cdot a \\&= a^{-1} \cdot a \\&= 1\end{aligned}$$

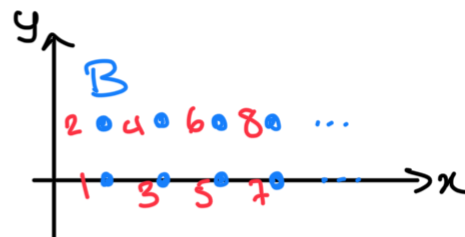
So $a^{-1} \cdot b^{-1}$ is the multiplicative inverse $(ab)^{-1}$ of ab . $\square$

2.) (This is a special case of HW2#1.)

(a.) $\exists f: A \rightarrow \mathbb{N}$ bijective

(b.) Define $f: B \rightarrow \mathbb{N}$ by

$$f(x, y) = \begin{cases} 2x-1 & \text{if } y=0 \\ 2x & \text{if } y=1 \end{cases}$$



Want: f bijective. Consider $g: \mathbb{N} \rightarrow B$,

$$g(n) = \begin{cases} (\frac{n+1}{2}, 0) & \text{if } n \text{ odd} \\ (\frac{n}{2}, 1) & \text{if } n \text{ even} \end{cases}$$

Then

$$(f \circ g)(n) = \begin{cases} f(\frac{n+1}{2}, 0) = 2(\frac{n+1}{2}) - 1 = n & \text{if } n \text{ odd} \\ f(\frac{n}{2}, 1) = 2(\frac{n}{2}) = n & \text{if } n \text{ even} \end{cases}$$

$$(g \circ f)(x, y) = \begin{cases} g(2x-1) = (\frac{(2x-1)+1}{2}, 0) = (x, 0) & \text{if } y=0 \end{cases}$$

$$g(2x) = \left(\frac{2x}{2}, 1\right) = (x, 1) \quad \text{if } y=1$$

for all $n \in \mathbb{N}$ and $(x, y) \in B$. So f is bijective by HW. $\square$

3.) (This is HW3 #5.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A, \sup B \in \mathbb{R}$ exist.

① $\sup A + \sup B$ is an upper bound for S :

As $\sup A, \sup B$ are upper bounds for A, B ,

$$a \leq \sup A \quad \forall a \in A, \quad b \leq \sup B \quad \forall b \in B$$

$$\Rightarrow a + b \leq \sup A + \sup B \quad \forall a \in A, \forall b \in B$$

② L upper bound for $S \Rightarrow L \geq \sup A + \sup B$:

Suppose not. Then L is an upper bound for S , but $L < \sup A + \sup B$. Set

$$\varepsilon = \frac{1}{2} (\sup A + \sup B - L) > 0.$$

As $\sup A - \varepsilon < \sup A$, then $\exists a_0 \in A$ s.t.

$$a_0 > \sup A - \varepsilon.$$

As $\sup B - \varepsilon < \sup B$, then $\exists b_0 \in B$ s.t.

$$b_0 > \sup B - \varepsilon.$$

Together,

$$a_0 + b_0 > \sup A + \sup B - 2\varepsilon = L.$$

This contradicts that L is an upper bound for S .

Collecting ① and ②, we conclude $\sup S$

$$= \sup A + \sup B.$$

□

4.) (This is HW4 #1.)

(a.) Prove: Fix $\varepsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$, $\exists N \in \mathbb{N}$
s.t. $|a_n - a| < \varepsilon \quad \forall n \geq N$

Then for $n \geq N$ we have

$$||a_n| - |a|| \leq |a_n - a| < \varepsilon$$

So $\lim_{n \rightarrow \infty} |a_n| = |a|$.

□

(b.) Disprove: Consider $a_n = (-1)^n$. Then:

- $\{|a_n|\}_{n \geq 1}$ converges: since $|a_n| = 1 \quad \forall n \geq 1$.
- $\{a_n\}_{n \geq 1}$ diverges: by lecture

□

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1. (10 points) Using induction, prove that for all $n \in \mathbb{N}$ we have

$$1 + 5 + 9 + \cdots + (4n - 3) = 2n^2 - n.$$

2. Let $f : A \rightarrow B$ be a function. **Prove or disprove** the following statements:

(a) (5 points) For all $Y \subseteq B$, we have $f(f^{-1}(Y)) \subseteq Y$.

(b) (5 points) For all $Y \subseteq B$, we have $f(f^{-1}(Y)) \supseteq Y$.

3. (10 points) Let $A, B \subseteq \mathbb{R}$ be nonempty and bounded sets. Prove that if $a \leq b$ for all $a \in A$ and for all $b \in B$, then $\sup A \leq \inf B$.

-
4. (10 points) Prove that if $\{a_n\}_{n \geq 1}$ is a sequence of positive real numbers that converges to $a > 0$, then the sequence $\{\sqrt{a_n}\}_{n \geq 1}$ converges to $\sqrt{a}$.

Extra paper

1.) (Similar to HW1#1.)

Base case: For $n=1$, we have $1 = 2 \cdot 1^2 - 1$, which is true.

Inductive step: Assume

$$1 + 5 + \dots + (4n-3) = 2n^2 - n$$

for some $n \geq 1$. Adding $(4n+1)$ to both sides,

$$\begin{aligned} 1 + 5 + \dots + (4n-3) + (4n+1) &= 2n^2 - n + (4n+1) \\ &= 2(n^2 + 2n + 1) - n - 1 \\ &= 2(n+1)^2 - (n+1). \end{aligned}$$

So the formula holds for $n+1$.

By induction, we conclude the formula is true $\forall n \geq 1$. $\square$

2.) (Similar to HW1#3b)

(a.) Prove: Fix $Y \subseteq B$ and $y \in f(f^{-1}(Y))$.

Then $\exists x \in f^{-1}(Y)$ s.t. $f(x) = y$. As $x \in f^{-1}(Y)$, then $f(x) \in Y$, and so $y = f(x) \in Y$.

(b.) Disprove: Consider $f: \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2$ and $Y = \{-1, 1\}$. Then

$$f^{-1}(\{-1, 1\}) = \{-1, 1\}, \quad f(\{-1, 1\}) = \{1\}$$

and so

$$f(f^{-1}(\{-1, 1\})) = f(\{-1, 1\}) = \{1\} \neq \{-1, 1\}$$

(By HW1#3b, we just need to pick a function f that is not surjective.)

3.) (This is HW3#6a.)

As $A, B \subseteq \mathbb{R}$ are nonempty and bounded, $\sup A, \inf B \in \mathbb{R}$ exist.

Let $b \in B$. As $a \leq b \ \forall a \in A$, then b is an upper bound for A , and so

$$\sup A \leq b.$$

This is true $\forall b \in B$. So $\sup A$ is a lower bound for B , and therefore

$$\sup A \leq \inf B.$$

□

4.) (This is HW4#5.)

Fix $\varepsilon > 0$. As $\lim_{n \rightarrow \infty} a_n = a$ and $a > 0$, then $\exists N \in \mathbb{N}$ s.t.

$$|a_n - a| < \sqrt{a} \cdot \varepsilon \quad \forall n \geq N.$$

Then for $n \geq N$ we have

$$|\sqrt{a_n} - \sqrt{a}| = \left| \frac{(\sqrt{a_n} - \sqrt{a})(\sqrt{a_n} + \sqrt{a})}{\sqrt{a_n} + \sqrt{a}} \right| = \frac{|a_n - a|}{\sqrt{a_n} + \sqrt{a}}$$

$$< \frac{|a_n - a|}{\sqrt{a}} < \frac{\sqrt{a} \cdot \varepsilon}{\sqrt{a}} = \varepsilon$$

$\nearrow$
 $\sqrt{a_n} > 0$

So $\lim_{n \rightarrow \infty} \sqrt{a_n} = \sqrt{a}$.

□